Project 2 - Bayesian Networks

Bayesian Network for Stress Data (How Lifestyle Affects Stress)

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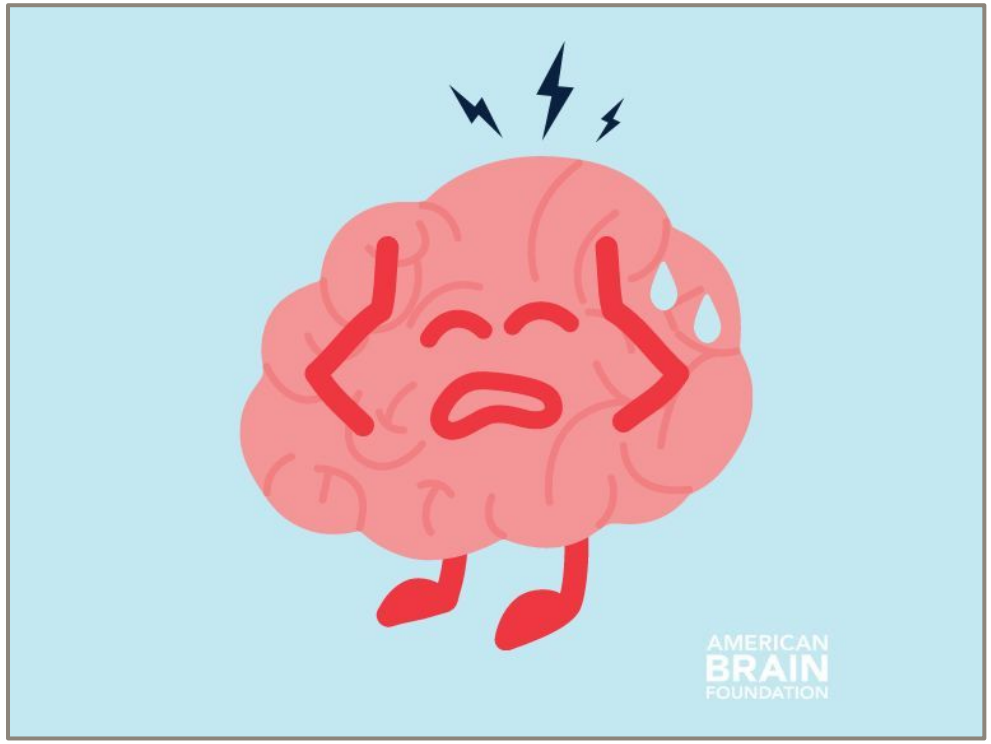
Github Source Code

Stress

- Life is inherently full of stressors, everything is interconnected
- These can have major underlying effects on our health

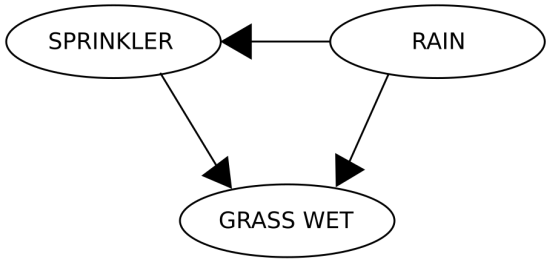


The Average Engineering Student's Brain 24/7

Bayesian Networks

- We can use data to rationalize and understand our lived experiences
- Bayesian Networks are a type of Probabilistic Graphical Model that can be used to:
  - Describe
  - Diagnose
  - Predict

	SPRINKLER	
RAIN	T	F
F	0.4	0.6
T	0.01	0.99



	RAIN	
T	F	
0.2	0.8	

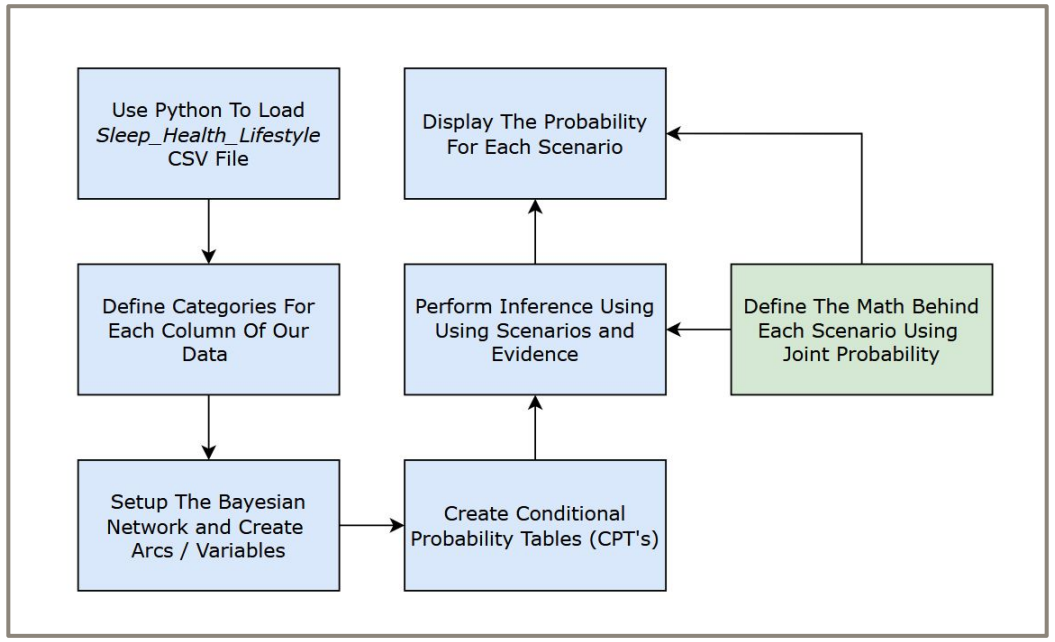
		GRASS WET	
SPRINKLER	RAIN	T	F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

An Example Bayesian Network

Procedure and Flowchart

Objectives

- Build a **Bayesian Network** using real world data
  - Age
  - Occupation
  - Sleep
  - Physical Health
- Compute **Conditional Probability Tables (CPTs)**
- Analyze how lifestyle factors influence stress
- Perform **inference using different scenarios**



An Example Bayesian Network

Bayesian Network Setup

```python
# Define BN
bn = gum.BayesNet('DynamicStressBN')

# Add Variables Dynamically
add_variable(bn, df, INPUT_1)
add_variable(bn, df, INPUT_2)
add_variable(bn, df, TARGET)
add_variable(bn, df, OUTPUT_1)
add_variable(bn, df, OUTPUT_2)

# Add Arcs Dynamically
bn.addArc(INPUT_1, TARGET)
bn.addArc(INPUT_2, TARGET)
bn.addArc(TARGET, OUTPUT_1)
bn.addArc(TARGET, OUTPUT_2)

# Show BN
gnb.showBN(bn)
```

```python
# User Configuration
INPUT_1 = 'Occupation'
INPUT_2 = 'Demographic'
TARGET = 'Stress Level'
OUTPUT_1 = 'Heart Rate'
OUTPUT_2 = 'Quality of Sleep'
SELECTED_OCCUPATIONS = ['Doctor', 'Engineer', 'Teacher']
```

```python
# Compute Prior Probabilities
prior_1 = compute_prior(df, INPUT_1)
prior_2 = compute_prior(df, INPUT_2)

# Compute Target Conditional Probability
target_counts = compute_conditional(df, [INPUT_1, INPUT_2], TARGET)

# Compute Output Conditional Probabilities
output1_counts = compute_conditional(df, [TARGET], OUTPUT_1)
output2_counts = compute_conditional(df, [TARGET], OUTPUT_2)
```

Bayesian Network Setup

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Add Variable

- Build the node with the appropriate name and ordered states, and then adds that new node into the network structure

Compute Prior

- Counts how many times each unique value appears in the column and divides by the total row count
- Reformats it into a clean, two column dataframe

Compute Conditional

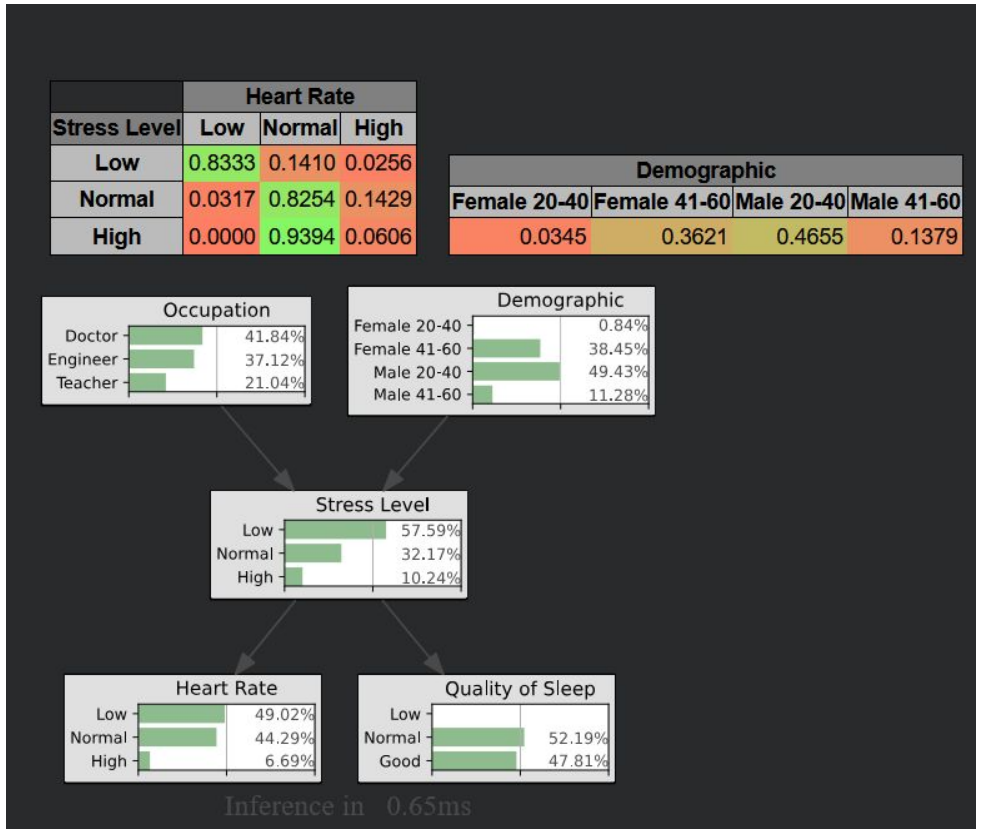
- Groups the dataset by the parent and child columns for how often specific combinations happen together
- Divide the count of each specific combination by the total occurrences of the parent state

Fill Output CPT

- Loops through every possible state of the parent variable of *counts* dataframe to isolate the probabilities for the child variable under that specific parent state.

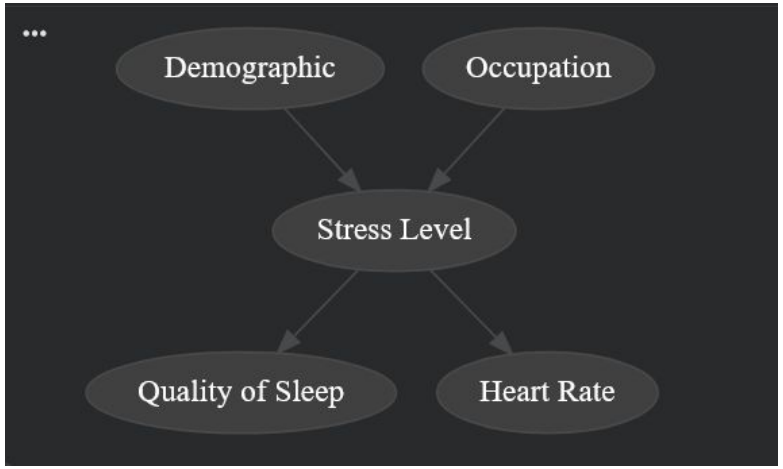
```python
# Prior Probability Function
def compute_prior(df, var):
    counts = df[var].value_counts(normalize=True).reset_index()
    counts.columns = [var, 'Probability']
    return counts

# Conditional Probability Function
def compute_conditional(df, parents, child):
    group_cols = parents + [child]
    counts = df.groupby(group_cols).size().reset_index(name='count')

    counts['Probability'] = counts.groupby(parents)['count'].transform(
        lambda x: x / x.sum()
    )
    return counts

# Add Variables Dynamically
def add_variable(bn, df, var):
    if var in STATE_ORDERS:
        states = STATE_ORDERS[var]
    else:
        states = sorted(df[var].dropna().unique())

    bn.add(gum.LabelizedVariable(var, '', states))
```

Bayesian Network Setup



	Heart Rate		
Stress Level	Low	Normal	High
Low	0.8333	0.1410	0.0256
Normal	0.0317	0.8254	0.1429
High	0.0000	0.9394	0.0606

Demographic			
Female 20-40	Female 41-60	Male 20-40	Male 41-60
0.0345	0.3621	0.4655	0.1379

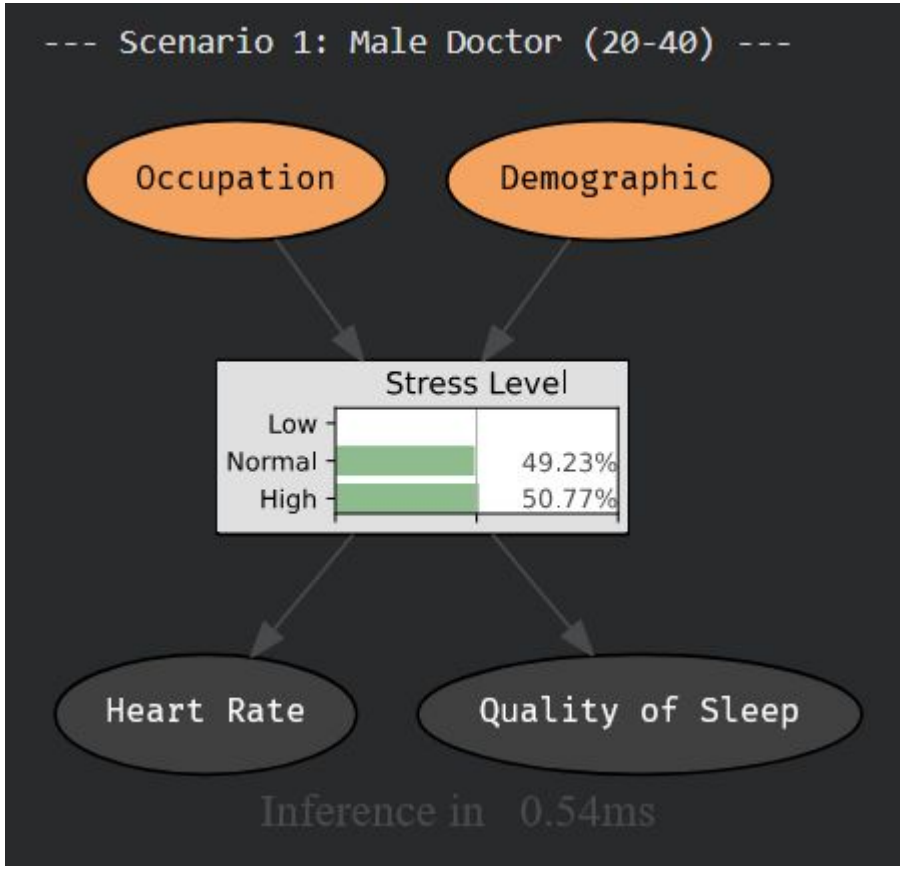


		Stress Level		
Demographic	Occupation	Low	Normal	High
Female 20-40	Doctor	0.0000	0.0000	0.0000
	Engineer	0.0000	0.0000	0.0000
	Teacher	1.0000	0.0000	0.0000
Female 41-60	Doctor	1.0000	0.0000	0.0000
	Engineer	1.0000	0.0000	0.0000
	Teacher	0.8621	0.1379	0.0000
Male 20-40	Doctor	0.0000	0.4923	0.5077
	Engineer	0.9091	0.0909	0.0000
	Teacher	0.0000	1.0000	0.0000
Male 41-60	Doctor	0.5000	0.5000	0.0000
	Engineer	0.0500	0.9500	0.0000
	Teacher	0.0000	0.0000	0.0000

	Quality of Sleep		
Stress Level	Low	Normal	Good
Low	0.0000	0.3205	0.6795
Normal	0.0000	0.7302	0.2698
High	0.0000	1.0000	0.0000

Occupation		
Doctor	Engineer	Teacher
0.4080	0.3621	0.2299

Scenario 1: Normal Stress for a 20-40 year old Doctor



1) $P(S=\text{Normal} \mid O=\text{Doctor}, \text{Demographic}=\text{Male } 20\text{-}40)$

$P(A|B) = \frac{P(A,B)}{P(B)} \xrightarrow{3\ \text{Var}}$ $P(A,B,C) = P(A|B,C)P(B,C)$

$P(B,C) = P(B|C)P(C)$

$P(A,B,C) = P(A|B,C)P(B|C)P(C)$

$P(B|C) = P(B)$ if B is independent

Chain Rule!

$P(S|O,D) = \frac{P(S,O,D)}{P(O,D)} \longrightarrow P(S,O,D) = P(S|O,D)P(O)P(D) \longrightarrow P(O,D) = P(O)P(D)$

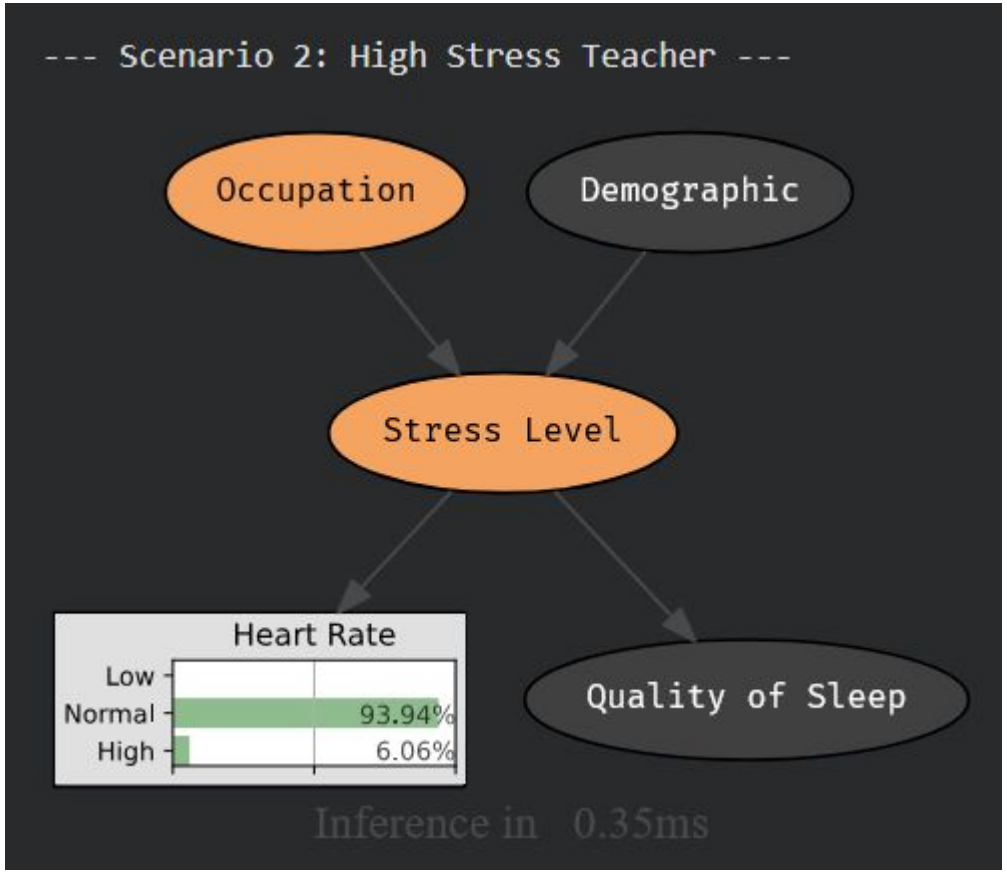
$P(S|O,D) = \frac{P(S|O,D)P(O)P(D)}{P(O)P(D)} \longrightarrow P(S|O,D) = \frac{0.4923 \times 0.408 \times 0.4655}{0.408 \times 0.4655} = \frac{0.0935}{0.1899} \longrightarrow P(S|O,D) = 0.4923$

$P(S=\text{Normal} \mid O=\text{Doctor}, \text{Demographic}=\text{Male } 20\text{-}40) = 0.4923$

Probability Stress is Normal given Occupation is Doctor and Demographic is Male 20-40

		Stress Level		
Demographic	Occupation	Low	Normal	High
Female 20-40	Doctor	0.0000	0.0000	0.0000
	Engineer	0.0000	0.0000	0.0000
	Teacher	1.0000	0.0000	0.0000
Female 41-60	Doctor	1.0000	0.0000	0.0000
	Engineer	1.0000	0.0000	0.0000
	Teacher	0.8621	0.1379	0.0000
Male 20-40	Doctor	0.0000	0.4923	0.5077
	Engineer	0.9091	0.0909	0.0000
	Teacher	0.0000	1.0000	0.0000
Male 41-60	Doctor	0.5000	0.5000	0.0000
	Engineer	0.0500	0.9500	0.0000
	Teacher	0.0000	0.0000	0.0000

Scenario 2: High Heart Rate for a High Stress Teacher



2) $P(HR = High \mid O = Teacher, S = High)$

$$P(HR \mid O,S) = \frac{P(HR,O,S)}{P(O,S)} \longrightarrow \frac{P(HR|S)\cancel{P(O)}\cancel{\sum_D P(S|O,D)P(D)}}{\cancel{P(O)}\cancel{\sum_D P(S|O,D)P(D)}} \longrightarrow P(HR \mid O,S) = P(HR|S) = 0.060$$

$$P(HR,O,S) = \sum_D P(HR|S)P(S|O,D)P(O)P(D) = P(HR|S)P(O)\sum_D P(S|O,D)P(D)$$

$$P(O,S) = \sum_D P(S|O,D)P(O)P(D) = P(O)\sum_D P(S|O,D)P(D)$$

$$P(HR = High \mid O = Teacher, S = High) = 0.060$$

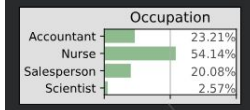
Probability Heart Rate is High given Occupation is Teacher and Stress is High

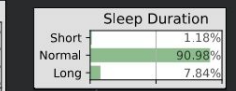
	Heart Rate		
Stress Level	Low	Normal	High
Low	0.8333	0.1410	0.0256
Normal	0.0317	0.8254	0.1429
High	0.0000	0.9394	0.0606

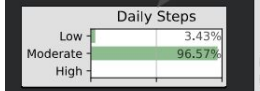
(New) Bayesian Network Setup

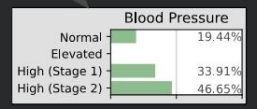
		Stress Level		
Sleep Duration	Occupation	Low	Normal	High
Short	Accountant	0.0000	0.0000	0.0000
	Nurse	0.0000	0.0000	1.0000
	Salesperson	0.0000	0.0000	0.0000
	Scientist	0.0000	0.0000	1.0000
Normal	Accountant	0.7568	0.2432	0.0000
	Nurse	0.2600	0.1200	0.6200
	Salesperson	0.0000	1.0000	0.0000
	Scientist	0.0000	1.0000	0.0000
Long	Accountant	0.0000	0.0000	0.0000
	Nurse	1.0000	0.0000	0.0000
	Salesperson	0.0000	0.0000	0.0000
	Scientist	0.0000	0.0000	0.0000

Occupation			
Accountant	Nurse	Salesperson	Scientist
0.2552	0.4966	0.2207	0.0276

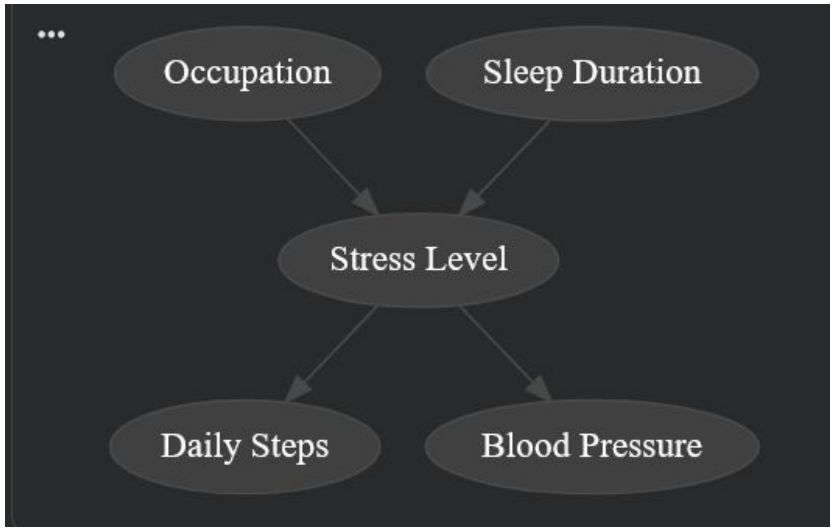








Inference in 0.60ms

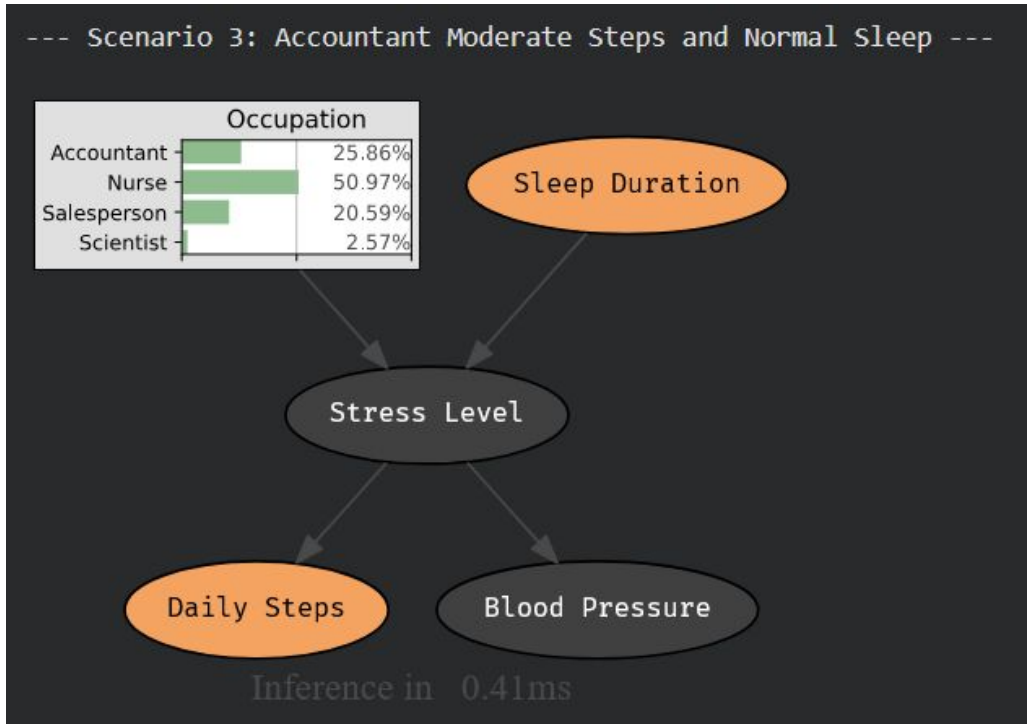


	Daily Steps		
Stress Level	Low	Moderate	High
Low	0.0000	1.0000	0.0000
Normal	0.1020	0.8980	0.0000
High	0.0000	1.0000	0.0000

Sleep Duration		
Short	Normal	Long
0.0207	0.8345	0.1448

	Blood Pressure			
Stress Level	Normal	Elevated	High (Stage 1)	High (Stage 2)
Low	0.4677	0.0000	0.0161	0.5161
Normal	0.0612	0.0000	0.9388	0.0000
High	0.0000	0.0000	0.0588	0.9412

(New) Scenario 1: Occupation given Moderate Daily Steps and Normal Sleep



Occupation			
Accountant	Nurse	Salesperson	Scientist
0.2552	0.4966	0.2207	0.0276

3) $P(O = \text{Accountant} \mid D = \text{Moderate}, SD = \text{Normal})$

$$P(O \mid D, SD) = \frac{P(O, D, SD)}{P(D, SD)}$$

$$P(O, D, S) = \sum_S P(O, SD, S, D) \longrightarrow P(O)P(SD)\sum_S P(S \mid O, SD)P(D \mid S)$$

$$P(O, SD, S, D) = P(O)P(SD)P(S \mid O, SD)P(D \mid S)$$

$$P(D, SD) = \sum_O [P(O)P(SD)\sum_S P(S \mid O, SD)P(D \mid S)]$$

$$P(O \mid D, SD) = \frac{P(O)\cancel{P(SD)}\sum_S P(S \mid O, SD)P(D \mid S)}{\sum_O [P(O)\cancel{P(SD)}\sum_S P(S \mid O, SD)P(D \mid S)]} = \frac{P(O)\sum_S P(S \mid O, SD)P(D \mid S)}{\sum_O [P(O)\sum_S P(S \mid O, SD)P(D \mid S)]}$$

Accountant When SD = Normal D = Moderate (Num)

$$\left.\begin{array}{l} S = \text{Low} \longrightarrow 0.7568 \times 1.000 = 0.7568 \\ S = \text{Normal} \longrightarrow 0.2432 \times 0.8980 = 0.2184 \\ S = \text{High} \longrightarrow 0.000 \times 1.000 = 0.00 \end{array}\right] \longrightarrow P(O)\sum_S P(S \mid O, SD)P(D \mid S) = 0.2552 \times (0.7568 + 0.2184 + 0.00) = 0.2489$$

Other Occupations (Denom)

Sum All

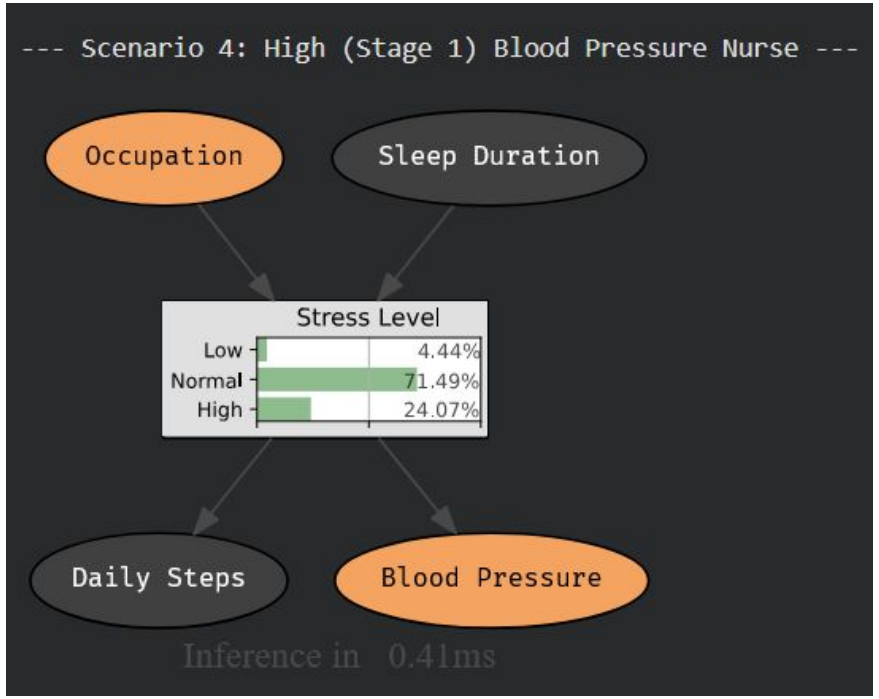
$$\left.\begin{array}{l} \text{Nurse} \longrightarrow 0.4966 \times [(0.26 \times 1) + (0.12 \times 0.898) + (0.62 \times 1)] = 0.4905 \\ \text{Salesperson} \rightarrow 0.2207 \times [(0.00 \times 1) + (1.0 \times 0.898) + (0.00 \times 1)] = 0.1982 \\ \text{Scientist} \longrightarrow 0.0276 \times [(0.00 \times 1) + (1.0 \times 0.898) + (0.00 \times 1)] = 0.0248 \end{array}\right] \sum_O [P(O)\sum_S P(S \mid O, SD)P(D \mid S)] = 0.9624$$

$$P(O = \text{Accountant} \mid D = \text{Moderate}, SD = \text{Normal}) = \frac{0.2489}{0.9624} = 0.2586$$

Probability <u>Occupation is Accountant</u> given

<u>Daily Steps is Moderate</u> and <u>Sleep Duration is Normal</u>

(New) Scenario 2 - Stress given a Stage 1 Blood Pressure Nurse



		Stress Level		
Sleep Duration	Occupation	Low	Normal	High
Short	Accountant	0.0000	0.0000	0.0000
	Nurse	0.0000	0.0000	1.0000
	Salesperson	0.0000	0.0000	0.0000
	Scientist	0.0000	0.0000	1.0000
Normal	Accountant	0.7568	0.2432	0.0000
	Nurse	0.2600	0.1200	0.6200
	Salesperson	0.0000	1.0000	0.0000
	Scientist	0.0000	1.0000	0.0000
Long	Accountant	0.0000	0.0000	0.0000
	Nurse	1.0000	0.0000	0.0000
	Salesperson	0.0000	0.0000	0.0000
	Scientist	0.0000	0.0000	0.0000

4) $P(S=\text{Low} \mid O=\text{Nurse}, B=\text{H1})$

$$P(S|O,B) = \frac{P(S,O,B)}{P(O,B)}$$

$$P(S,O,B,SD) = P(O)P(SD)P(S|O,SD)P(B|S)$$

$$P(S,O,B) = P(O)P(B|S)\sum_{SD} P(SD)P(S|O,SD)$$

$$P(O,B) = \sum_{S}[P(O)P(B|S)\sum_{SD} P(SD)P(S|O,SD)]$$

$$P(S|O,B) = \frac{\cancel{P(O)}P(B|S)\sum_{SD} P(SD)P(S|O,SD)}{\sum_{S}[\cancel{P(O)}P(B|S)\sum_{SD} P(SD)P(S|O,SD)]} = \frac{P(B|S)\sum_{SD} P(SD)P(S|O,SD)}{\sum_{S}[P(B|S)\sum_{SD} P(SD)P(S|O,SD)]}$$

$\sum_{SD}$ Block $O=\text{Nurse}$

$S=\text{Low} \longrightarrow (0.0207 \times 0) + (0.8345 \times 0.26) + (0.1448 \times 1) = 0.3618$
$S=\text{Normal} \longrightarrow (0.0207 \times 0) + (0.8345 \times 0.12) + (0.1448 \times 0) = 0.1001$
$S=\text{High} \longrightarrow (0.0207 \times 1) + (0.8345 \times 0.62) + (0.1448 \times 0) = 0.5381$

$B=\text{H1}$ $S=\text{Low}$ (Num)

$P(B=\text{H1}|S=\text{Low}) \times 0.3618 = 0.0161 \times 0.3618 = 0.0058 \longrightarrow P(B|S)\sum_{SD} P(SD)P(S|O,SD) = 0.0058$

$\sum_{S}$ Block (Denom)

Sum All

$S=\text{Low} \longrightarrow P(B=\text{H1}|S=\text{Low}) \times 0.3618 = 0.0161 \times 0.3618 = 0.0058$
$S=\text{Normal} \longrightarrow P(B=\text{H1}|S=\text{Normal}) \times 0.1001 = 0.9388 \times 0.1001 = 0.0940$
$S=\text{High} \longrightarrow P(B=\text{H1}|S=\text{High}) \times 0.5381 = 0.0588 \times 0.5381 = 0.0316$

$\sum_{S}[P(B|S)\sum_{SD} P(SD)P(S|O,SD)] = 0.1314$

$$P(S=\text{Low} \mid O=\text{Nurse}, B=\text{H1}) = \frac{0.0058}{0.1314} = 0.0441$$

Probability Stress is Low given Occupation is Nurse and Blood Pressure is High (Stage 1)

Conclusions

- Health and lifestyle are very interconnected
- Bayesian networks allow us to model these relationships
- In this investigation, we realized how much of an impact lifestyles have on stress and health
- Take care of yourself!

Limitations:

- The dataset we used could potentially have bias, for example, there were no doctors with high sleep scores
- We only had 5 nodes in our Bayesian network, with so many factors we can easily miss relationships in data

The Average Engineering Student After April 30th

Future Steps

- If we were to do this project again, we could look at additional variables, such as family structure, financial situation, or country of residence
- We could expand the network to account for more variables

Maslow's hierarchy of needs